

Motivation

- The number of athletes entering the transfer portal for NCAA Division I men's basketball has increased in recent years (>1,800 successful transfers in 2025).
- Transferring has a huge impact on coaches, players, and collegiate athletics programs.

Research question: What is the effect of transferring on a player's athletic performance in NCAA Division I men's basketball?

Data and Methods

Original data: Bart Torvik (accessed through hoopR package) → player statistics and team ratings by season (2022-23 to 2025-26)

Adding features: Player's transfer status, previous season's statistics, conference ratings (= average of all team ratings across the conference)

Propensity score matching: Matching players with similar position, experience, team and conference ratings → Avoid selection biases associated with transfers

Matched data

Unmatched data

Mixed effects regression:

$$PPG_{current} = a + \beta_1 \text{Transferred} + \beta_2 \text{Season} + \beta_3 PPG_{previous} + \beta_4 \text{Position} + \beta_5 \text{Experience} + \beta_6 \text{Conference} + \beta_7 \text{TeamRating} + \beta_8 \text{ConfRating}$$

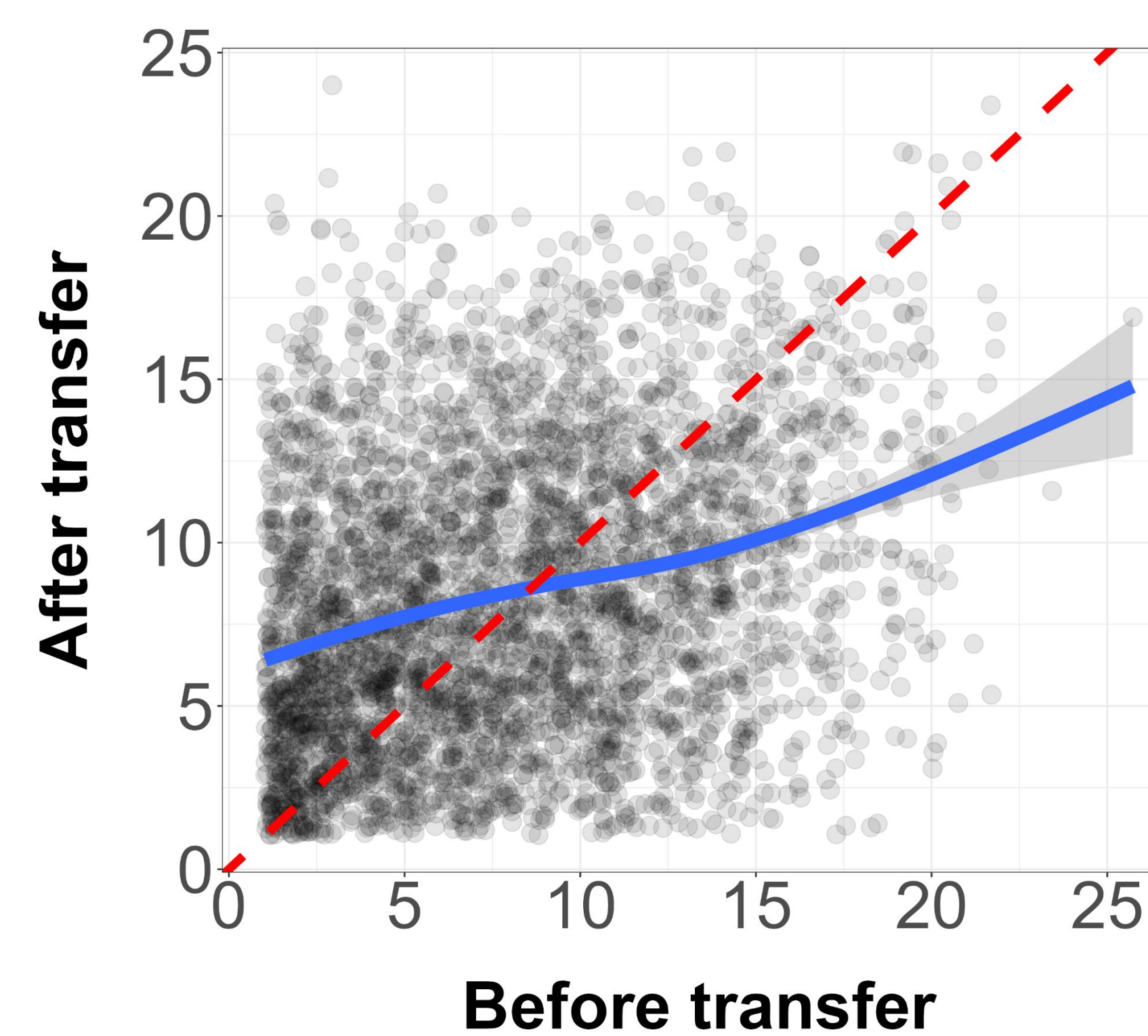
- Controlling for player's natural development along with team, conference, and season context
- **Random intercepts** for player-team interaction → controlling for different baseline performances

β_1 : Effect of interest

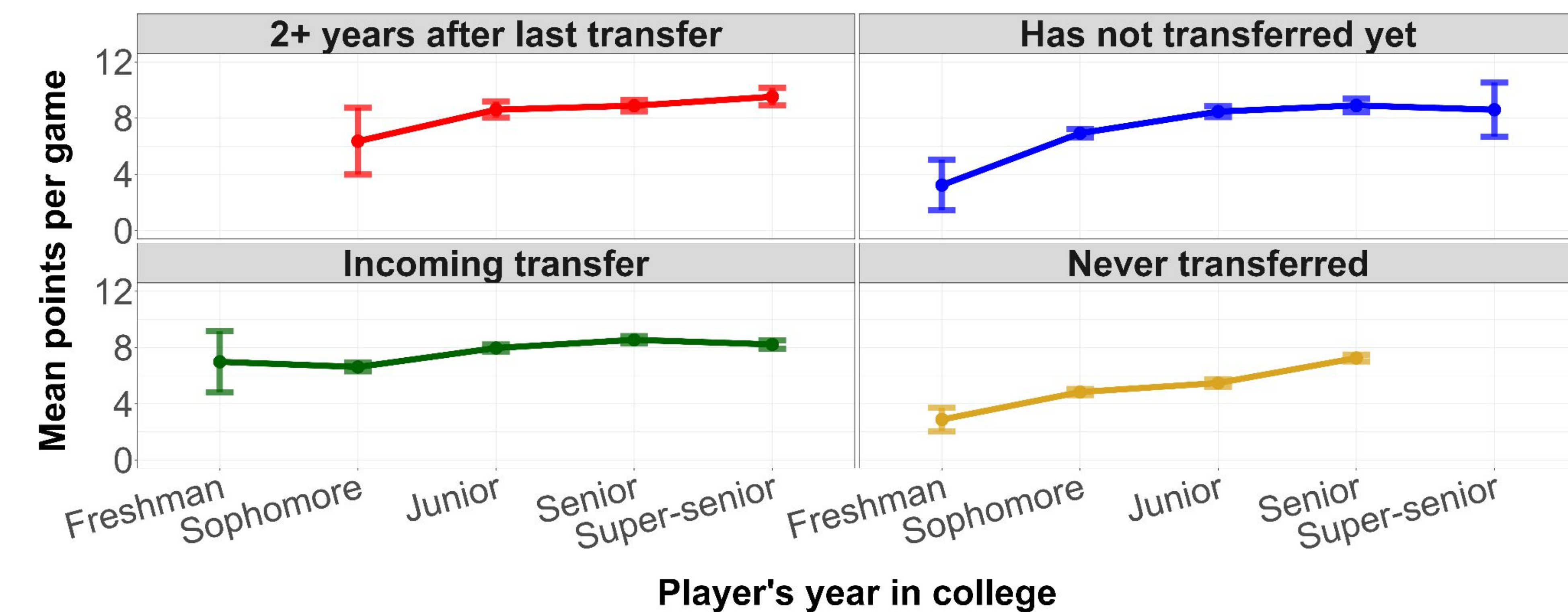
- Coefficient value for "Transferred" (indicator variable: 1 if incoming transfer, 0 otherwise)
- The **expected change in points per game** after a transfer. Standard errors included to quantify uncertainty

Results

General Trends in Points Per Game (PPG), Before and After Transfer

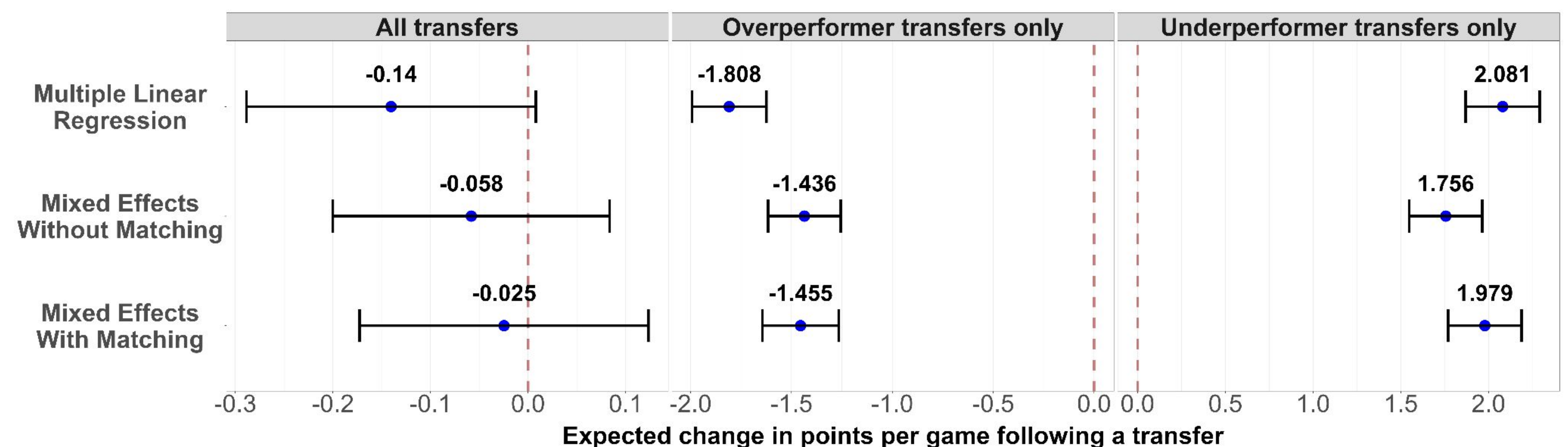


Mean Points Per Game Separated by Experience Level and Transfer Status



Error bars represent the corresponding 95% confidence intervals.

Effect of Transferring in General, as an Overperformer, and as an Underperformer (inspired by Cole et al., 2026)



Overperformers are players whose PPG is greater than or equal to the median PPG across all team members.

Underperformers are players whose PPG is below the median PPG across all team members.

Discussion

Recommendations:

- Coaches: Focus on **underperformers at bigger/more successful programs** in the transfer portal.
- Players: If getting success, **re-evaluate transfer decisions**.

Limitations:

- Some players have **complex transfer histories**.
- Frequent changes in **eligibility policies**.
- Transfers for non-athletic reasons **and NIL**.
- **More comprehensive metric(s)** for player performance.

Reference

Cole, K., Pearman, F., & Khanna, S. (2026). Big Fish or Little Pond? National Evidence on the Relationship Between Transferring and Athletic Performance Within Division I Men's College Basketball. *Journal of Issues in Intercollegiate Athletics*, 19(1). <https://doi.org/10.51221/sc.jiia.2026.19.1.15>

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